

IAS PRELIMS/MAINS

Ecology & Environment

A stylized illustration of a green city skyline with various skyscrapers, wind turbines, and a sun, set against a light green background with a globe-like shape.

A COMPREHENSIVE COVERAGE

MODULE - 32

SOLID WASTE- 2

- The indiscriminate disposal of plastics has become a major threat to the environment.
- In particular, the plastic carry bags are the biggest contributors of littered waste and every year millions of plastic bags end up in to the environment (soil, fresh water bodies and oceans) and it takes an average of one thousand years to decompose complete.



SOLID WASTE

INTENSITY OF THE PROBLEM

- At least 8 million tons of plastic end up in our oceans every year. If plastic pollution continues to rise then oceans will have more plastics than fish by 2050.
- India generates around 56 lakh tons of plastic waste annually. 15,000 tons of plastic waste is generated in the country every day, out of which 9000 tons is collected and processed, but 6000 tons of plastic waste is not being collected.
- 42% of the total plastic waste is packaging material. Around 10,890 tons of packaging waste is generated per day.



WHAT NOT TO DO WITH PLASTICS?

- Burning of plastics releases two **Persistent Organic Pollutants** which are highly carcinogenic and toxic, they are
 - ✓ Polychlorinated Dibenzo Para Dioxins (PCDDs) and
 - ✓ Polychlorinated Dibenzo Furans (PCDFs) released by burning of PVCs.
- Careless disposal of **plastic bags** **chokes drains** and blocks the porosity of the soil.
- Plastic **disturbs the soil microbe activity.**
- Plastic **deteriorates soil fertility** as it forms part of manure and **remains in the soil for years.**



REMEDY FOR PLASTIC WASTE

- Polyblend is a fine powder of recycled plastic waste is used to lay roads.
- Polyblend has water repellent proper so it increases the road life significantly.



1

SINGLE USE PLASTICS

- Any plastic material which is used once and then disposed right away.
 - ✓ It is that plastic material that is produced and designed to be thrown away after being used once.
 - ✓ Most commonly used single use plastics items are: cigarette butts, plastic bottles and caps, straws, food wrappers, carry bags and disposable cutlery.
 - ✓ Plastic packaging accounts for nearly half of all the plastic waste in India and much of it is thrown away just after its first use.
 - ✓ When plastic is discarded in landfills or environment, it can take upto a thousand years to decompose and will break down into smaller fragments called microplastics.



PROBLEMS OF PLASTICS IN INDIA

2

MICROPLASTICS

- These are tiny plastic particles. They end up in water bodies posing a threat to humans and aquatic life.
- ✓ Microbeads are a kind of microplastic, they are used as exfoliates in cleansers and toothpaste. Even though they are banned in US and Canada, they are still used in India.

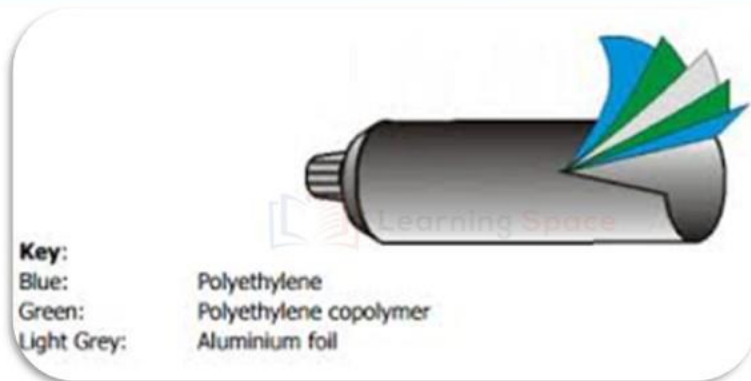


ECOLOGY AND ENVIRONMENT

PROBLEMS OF PLASTICS IN INDIA

3 MULTI-LAYERED-PACKAGING (MLP)

- Multi-layered plastics are made up of a layered composition of various types of plastics.
- ✓ MLP packets have multiple sheets of plastic enclosing a layer of aluminum. (Ex: wrapper of a biscuit packet).
- ✓ Technically MLP can be any material that has at least one layer of plastic.
- ✓ It stays in the environment forever because ragpickers do not pick them up. The government too does not have a clue how to deal with these packets which are indestructible and add to garbage dumps.
- ✓ A study found that 53 per cent of plastic waste in the cities was MLP



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Learning Space

For Suggestions:

suggestions@learningspace.in

For Information:

info.learningspacedigital@gmail.com

To Contact us:

info@learningspace.in

Visit us at:

www.learningspacedigital.com

98499 42299